SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0501
NAME : Suraj varma avulamanda
Reg.no: 192525339

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck	BL1 – Remember	5

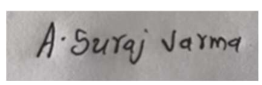
	deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5

4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. A department controls a number of PROJECT. Projects have a unique name, number and a single location. Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I Suraj Varma Avulamanda (192525339) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understandin g
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer

Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. UPS Product Tracking Information System – ER Diagram
Answer:
Entities and Attributes
1. Shipped_Item

- Item_Number (Primary Key)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Final_Delivery_Date

2. Retail_Center

- Unique_ID (Primary Key)
- Type
- Address

3. Transportation_Event

- Schedule_Number (Primary Key)
- Type (Flight/Truck)
- Delivery_Route

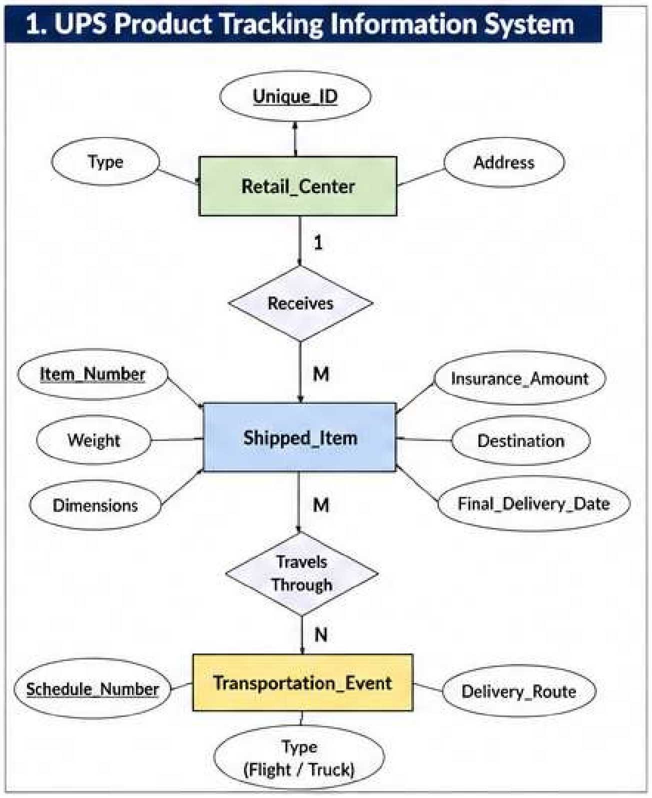
Relationships

- **Retail_Center** receives **Shipped_Item** (1:M)
- **Shipped_Item** travels through **Transportation_Event** (M:N)

ER Diagram (Text Representation)

```
Retail_Center
--------------
Unique_ID (PK)
Type
Address
    |
    | Receives (1:M)
    |
Shipped_Item
--------------
Item_Number (PK)
Weight
Dimensions
Insurance_Amount
Destination
Final_Delivery_Date
    |
    | Travels Through (M:N)
    |
Transportation_Event
---------------------
Schedule_Number (PK)
```

Type
Delivery_Route



Q2. Course Management System – ER Diagram

Answer:

Entities

Course

- Course_Number (PK)
- Name
- Department

Section (Weak Entity)

- Section_Number (Partial Key)
- Enrollment

Exam

- Exam_ID (PK)
- Time

Room

- Room_Number (PK)
- Capacity
- Building

Relationships

- Course → Has → Section (1:M)
- Section → Has → Exam (1:1)

- Exam → Conducted_In → Room (M:1)

ER Diagram

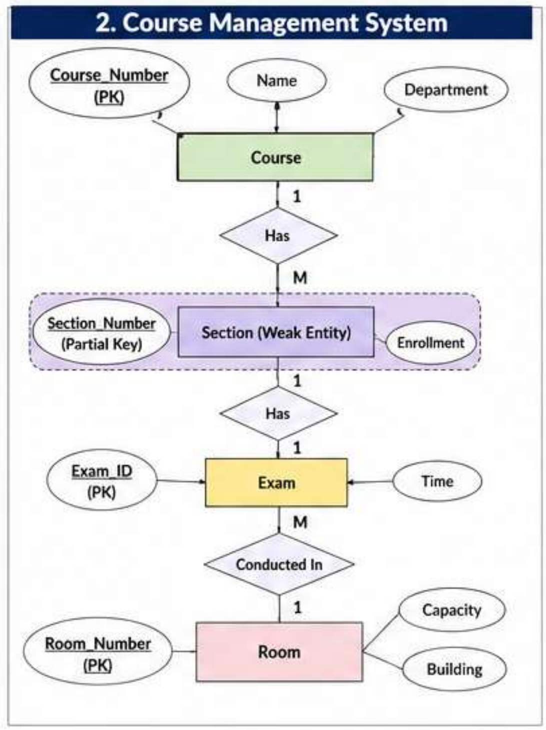
Course

Course_No (PK)
Name
Department
|
| Has (1:M)
|
Section (Weak Entity)

Section_No
Enrollment
|
| Has
|
Exam

Exam_ID
Time
|
| Conducted In
|
Room

Room_No (PK)
Capacity
Building



Q3. University Registrar's Office Management System

Answer:

Entities

Course

- Course_No (PK)
- Title
- Credits
- Syllabus
- Prerequisites

Course_Offering

- Offering_ID (PK)
- Year
- Semester

- Section_No
- Timings
- Classroom

Student

- Student_ID (PK)
- Name
- Program

Instructor

- Instructor_ID (PK)
- Name
- Department
- Title

Relationships

- Course → Offered_As → Course_Offering
- Instructor → Teaches → Course_Offering
- Student → Enrolls → Course_Offering
- Enrollment stores Grade

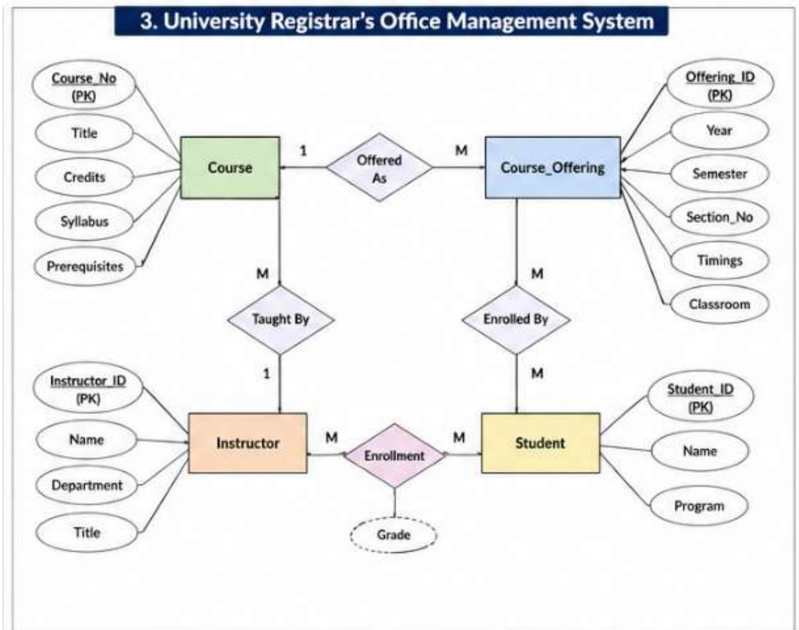
ER Diagram

```
Course
--------
Course_No
Title
Credits
Syllabus
Prerequisites
    |
    | Offered As
    |
Course_Offering
----------------
Offering_ID
Year
Semester
Section_No
Timings
Classroom
    |             |
    | Taught By       | Enrolled By
    |             |
Instructor       Student
-----------      ----------
Instructor_ID      Student_ID
Name             Name
Department         Program
```

Title

Enrollment

Grade



Q4. Hospital Management System
Answer:
Entities
Patient

- SS# (PK)
- Name
- Insurance

Physician

- Physician_Name (PK)
- Specialization

Test_Log

- SS#
- Test_Name
- Date
- Time

Patient_History

- SS#
- Test_Name
- Date

Relationship
Doctor treats Patient (M:N)

Tables

Patient

SS# (PK)
Name
Insurance

Physician

Physician_Name (PK)
Specialization

Doctor_Patient

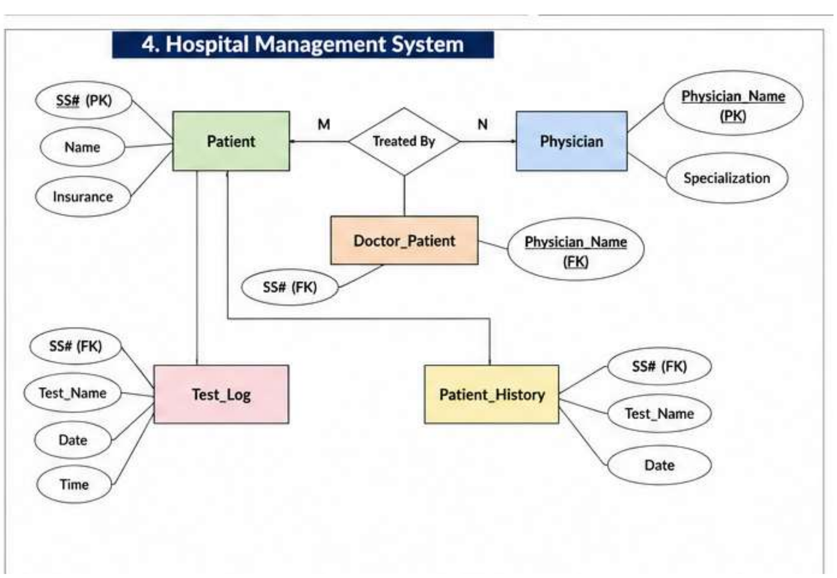
Physician_Name (FK)
SS# (FK)

Test_Log

SS#
Test_Name
Date
Time

Patient_History

SS#
Test_Name
Date



Q5. Company Project Management System
Answer:
Entities
Department

- Dept_ID (PK)
- Name

Employee

- SSN (PK)
- Name
- Address
- Salary
- Sex
- Birth_Date

Project

- Project_No (PK)
- Project_Name
- Location

Dependent

- Dependent_Name
- Birth_Date
- Relationship

Relationships

- Department manages Employee
- Department controls Project
- Employee works on Project
- Employee supervises Employee
- Employee has Dependents

ER Diagram

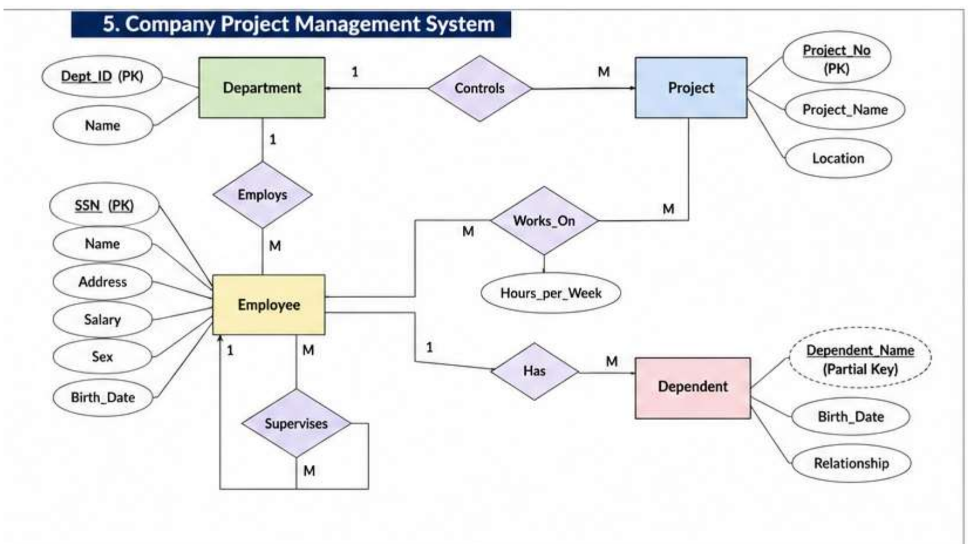
```
Department
------------
Dept_ID
Name
     |
     | Controls
     |
Project
---------
Project_No
Project_Name
Location

Department
     |
     | Employs
```

|
Employee

SSN
Name
Address
Salary
Sex
Birth_Date
|
| Has
|
Dependent

Dependent_Name
Birth_Date
Relationship



Q6. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.
Answer
Schema
A **schema** is the overall design or blueprint of a database. It defines the structure of the database, including tables, attributes, relationships, constraints, and data types. A schema changes very rarely.
Example:
STUDENT

Roll_No	Name	Age	Course

Here, the table structure is the **schema**.

Instance

An **instance** is the actual data stored in the database at a particular moment. It changes whenever records are inserted, updated, or deleted.

Example:

Roll_No	Name	Age	Course
101	Ramesh	20	CSE
102	Sita	21	ECE
103	Anil	19	ME

This data is called the **database instance**.

Difference between Logical, Physical, and View Schema

Logical Schema	Physical Schema	View Schema
Describes the overall logical structure of the database.	Describes how data is physically stored on the disk.	Describes only the portion of the database visible to a user.
Independent of hardware.	Depends on storage devices and indexing.	Used for security and simplicity.
Example: Tables, relationships, constraints.	Example: File organization, indexing.	Example: Student view, Faculty view.

Q7. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Answer

Entities and Attributes

1. Customer

- Customer_ID (Primary Key)
- Name
- Address
- Phone
- Email

2. Account

- Account_No (Primary Key)
- Account_Type
- Balance
- Opening_Date

3. Branch

- Branch_ID (Primary Key)
- Branch_Name
- Address

- City

4. Loan

- Loan_ID (Primary Key)
- Loan_Type
- Amount
- Status

5. Employee

- Employee_ID (Primary Key)
- Name
- Designation
- Salary

Relationships

- Customer **opens** Account (1:M)
- Customer **takes** Loan (1:M)
- Branch **maintains** Account (1:M)
- Branch **employs** Employee (1:M)
- Employee **manages** Branch (1:1)

Q8. Explain the Extended ER (EER) Model. Describe the concepts of Generalization, Specialization, and Aggregation.

Answer

Extended ER (EER) Model

The **Extended Entity Relationship (EER) Model** is an extension of the ER model that includes additional concepts such as inheritance, subclasses, superclasses, specialization, generalization, and aggregation. It is used to model complex real-world applications more effectively.

1. Generalization

Generalization is a **bottom-up** approach that combines two or more lower-level entities having common attributes into a higher-level entity (superclass).

Example:

Manager, Engineer, and Clerk → Employee

2. Specialization

Specialization is a **top-down** approach where one higher-level entity is divided into multiple lower-level entities based on their characteristics.

Example:

Employee → Manager, Engineer, Clerk

3. Aggregation

Aggregation is a technique in which a relationship set is treated as a higher-level entity so that it can participate in another relationship.

Example:
Instructor teaches Course, and the Teaching relationship is associated with a Department.

Advantages of EER Model

- Supports inheritance.
- Represents complex relationships.
- Reduces redundancy.
- Improves database design.

Q9. Design an EER Diagram for a Hospital Management System incorporating inheritance and weak entities.
Answer
Superclass
Person

- Person_ID (Primary Key)
- Name
- Address
- Phone

Subclasses (Inheritance)
Patient

- Patient_ID
- Insurance_No

Doctor

- Doctor_ID
- Qualification

Weak Entity
Test

- Test_Name *(Partial Key)*
- Test_Date
- Result

The **Test** entity depends on the **Appointment** entity for its identification, so it is considered a **weak entity**.

Other Entity
Appointment

- Appointment_ID
- Appointment_Date
- Time

Relationships

- Person **ISA** Patient or Doctor.

- Patient **has** Appointment (1:M).
- Appointment **includes** Test (1:M).
- Doctor **conducts** Test (1:M).

Features Used
- Inheritance (ISA)
- Weak Entity
- One-to-Many Relationships
- Primary and Partial Keys

Q10. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.
Answer
Database Administrator (DBA)
A **Database Administrator (DBA)** is responsible for managing, maintaining, securing, and ensuring the efficient operation of a database system.

Responsibilities of a DBA
1. **Schema Definition**
    - Creates database schema, tables, constraints, and indexes.
2. **Storage Management**
    - Manages storage structures, files, and indexing techniques.
3. **Security Management**
    - Creates user accounts, grants privileges, and protects database security.
4. **Backup and Recovery**
    - Performs regular backups and restores data after failures.
5. **Performance Tuning**
    - Optimizes SQL queries and improves database performance.
6. **Concurrency Control**
    - Manages multiple users accessing the database simultaneously.
7. **Integrity Enforcement**
    - Maintains data accuracy and consistency using constraints.
8. **Disaster Recovery Planning**
    - Prepares recovery plans for system failures and disasters.
9. **Monitoring and Maintenance**
    - Monitors database health, fixes issues, and performs routine maintenance.